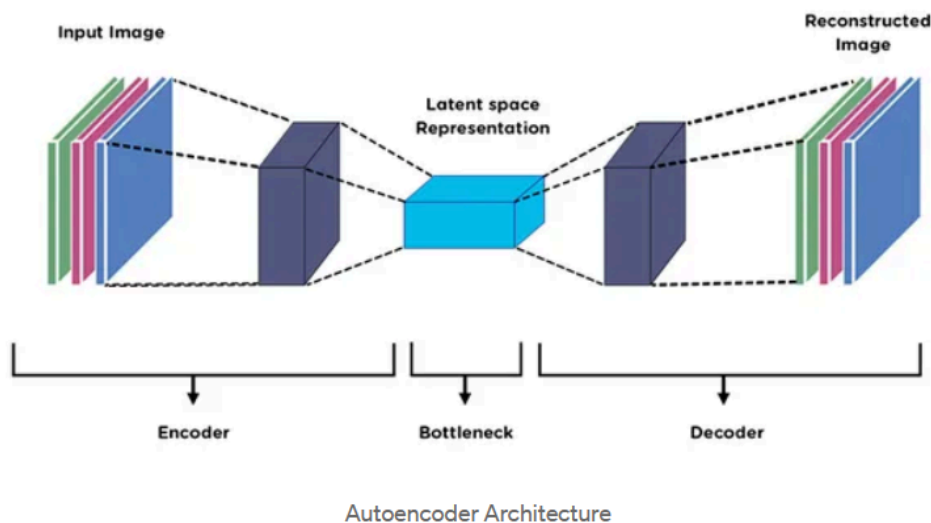


Class Activity based on GAN

1 What is an Autoencoder?

An Autoencoder is an Unsupervised artificial neural network designed to learn efficient data encodings by compressing input into a lower-dimensional “latent space” (encodings) and then reconstructing the original input(decoding).

2 Draw the Architecture of the Autoencoder?



An autoencoder network is composed of two parts **Encoder** and the **Decoder**.

Encoder: This part of the network encodes or compresses the input data into a latent-space representation. The compressed data typically looks garbled, nothing like the original data.

Decoder: This part of the network decodes or reconstructs the encoded data(latent space representation) back to the original dimension. The decoded data is a lossy reconstruction of the original data.

Idea behind Autoencoders:

Data compression is a big topic that's used in computer vision, computer networks, computer architecture, and many other fields. The point of data compression is to convert our input into a

smaller(Latent Space) representation that we recreate, to a degree of quality. This smaller representation is what would be passed around, and when anyone needed the original, they would reconstruct it from the smaller representation. And autoencoders are the networks that can be used for such tasks.

2 What is a Variable Autoencoder (VAE)?

Variational autoencoders (VAEs) are generative models used in machine learning (ML) to generate new data in the form of variations of the input data they're trained on. In addition to this, they also perform tasks common to other autoencoders, such as denoising. Like all autoencoders, variational autoencoders are deep learning models composed of an encoder that learns to isolate the important latent variables from training data and a decoder that then uses those latent variables to reconstruct the input data.

However, whereas most autoencoder architectures encode a discrete, fixed representation of latent variables, VAEs encode a **continuous, probabilistic representation of that latent space**.

This enables a VAE to not only accurately reconstruct the exact original input, but also use variational inference to generate new data samples that resemble the original input data.

Common examples include generating novel face images, creating fashion designs, or reconstructing missing parts of images

3 Compare an Autoencoder with Variational Autoencoder

Feature	Autoencoder (AE)	Variational Autoencoder (VAE)
Core Goal	Data compression & Feature extraction.	Data generation & Learning a smooth latent space.
Latent Space	Discrete/Deterministic: Maps input to a single fixed point in the latent vector.	Continuous/Probabilistic: Maps input to a distribution (mean μ and variance σ).
Bottleneck Output	A single vector (z).	Two vectors: Mean (μ) and Standard Deviation (σ).
Loss Function	Reconstruction Loss: Measures the difference between input and output.	Reconstruction Loss + KL Divergence: Ensures the latent space is Gaussian.
Generative Ability	Poor. The space between points is often "dead," leading to "garbage" output.	Excellent. The latent space is continuous, allowing for smooth interpolation.
Architecture	Simple Encoder-Decoder bottleneck.	Encoder-Decoder + Reparameterization Trick.

